Department of Civil & Environmental Engineering
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Morteza Shafiei

Ph.D. Candidate & University Instructor · Geotechnical Engineering

RESEARCH INTERESTS

Offshore geotechnics and site investigation · in-situ testing (CPTu) in crushable glauconitic sands · pile setup and deep-foundation mechanics · calibration-chamber physical modeling · particle morphology of granular materials.

EDUCATION

Ph.D., Civil & Environmental Engineering (Geotechnical Engineering) — 2021 – present

University of Massachusetts Amherst, Amherst, MA

Ph.D. candidate. Advisers: Prof. Don J. DeGroot, Sc.D., P.E., and Prof. Zachary J. Westgate, Ph.D., P.E.

M.Sc., Civil & Geotechnical Engineering — 2015 – 2018

Amirkabir University of Technology (Tehran Polytechnic)

Adviser: Prof. Kazem Fakharian. Graduated with highest academic honors. Designed and operated a large-scale pressurized consolidation calibration chamber for physical modeling of deep foundations.

B.Sc., Civil Engineering — 2010 – 2015

Solid mechanics, structural analysis, foundation engineering, and experimental soil mechanics.

PEER-REVIEWED JOURNAL PUBLICATIONS

1. Westgate, Z. J., Senanayake, A., **Shafiei, M.**, & DeGroot, D. J. (2026). Mitigation of CPT thrust force through glauconite sand. *Canadian Geotechnical Journal*. doi:10.1139/cgj-2026-0117
2. Kaviani-Hamedani, F., Esmailzade, M., Adineh, K., **Shafiei, M.**, & Shirkavand, D. (2024). Quantifying three-dimensional sphericity indices of irregular fine particles from 2D images through sequential sieving tests. *Granular Matter*, 26(1), Article 10. doi:10.1007/s10035-023-01380-z
3. Fakharian, K., **Shafiei, M.**, & Hafezan, S. (2023). Investigation of soil setup effects on pile response in clay considering overconsolidation ratio and installation method through physical modeling. *Canadian Geotechnical Journal*, 60(4), 488–503. doi:10.1139/cgj-2021-0686

THESIS & CONFERENCE RESEARCH

1. **Shafiei, M.**, & Fakharian, K. (2018). Effect of installation method and pile type on soil setup using consolidation chamber physical modeling. M.Sc. thesis and conference proceedings, Amirkabir University of Technology.

TEACHING & RESEARCH APPOINTMENTS

University Instructor

Department of Civil & Environmental Engineering, University of Massachusetts Amherst

Lead laboratory and course instructor for undergraduate engineering mechanics and geotechnical engineering courses.

Graduate Research Assistant

Offshore Geotechnics & Foundation Mechanics Group, University of Massachusetts Amherst

Full-scale field CPTu testing in glauconitic sands within the Piling in Glauconitic Sand (PIGS) Joint Industry Project.

SELECTED RESEARCH PROJECTS

CPTu thrust mitigation in glauconite sand — PIGS Joint Industry Project

- Field evaluation of six CPTu tooling configurations in the Hornerstown and Navesink glauconitic formations.
- 48 mm friction reducers with upper-bevel water jetting (6–7 L/min) extended penetration from ≈14 m to 23.5 m; standard friction reducers reduced rod thrust by 50–80%.

Calibration-chamber modeling of pile setup in clay

- Pressurized consolidation chamber and miniature model pile instrumented with 3 piezometers and 3 strain gauges.
- 21 static load tests (1 hour to 90 days) at OCR 1.2 and 4.8; separated EPWP dissipation (+287% in < 3 days) from thixotropic aging (+28.8% beyond 3 days).

3D particle sphericity from 2D images via sequential sieving

- Empirical relations linking 2D projection circularity to 3D sphericity and Zingg ratios, validated with X-ray micro-CT.

TECHNICAL SKILLS

Standards	ASTM D5778 (CPTu), ASTM D1143 (static pile load test), ASTM D6913 / D422 (sieve analysis), ASTM D2435 (consolidation), API RP 2GEO, ISO 19901-8
Instrumentation	Pressurized calibration chambers, miniature pore-pressure transducers, full-bridge strain gauges, CPTu friction reducers and water jetting, 3D X-ray micro-CT; sensor calibration and signal conditioning
Software	PLAXIS 2D & 3D, FLAC, GeoStudio, Python (NumPy, SciPy, pandas), MATLAB, ImageJ, gINT, CPT-Pro